

A Comparative Study of ARIMA, Double Moving Average, Random Forest, and Neural Network Autoregression Models for CNY/IDR Exchange Rate Forecasting

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Abstract—The Chinese yuan has become increasingly important to Indonesia as bilateral trade with China expands and a growing share of transactions is settled directly in local currencies under the Indonesia-China Local Currency Transaction framework. Accurate forecasting of the CNY/IDR exchange rate is therefore valuable for trade planning and currency risk management, yet the pair remains far less studied than IDR/USD. To address this gap, the present study compares five forecasting approaches: Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Double Moving Average (DMA), Random Forest, and Neural Network Autoregression (NNAR). The models were applied to daily CNY/IDR closing rates recorded from January 2024 to May 2026, giving 630 observations that were divided chronologically into 80% training and 20% testing data. Stationarity testing showed that the series was non-stationary in mean and became stationary after first-order differencing. Forecasting accuracy was evaluated using RMSE, MAE, and MAPE on the testing set. The Double Moving Average with a 24-day window achieved the best performance (RMSE 64.025, MAE 49.929, MAPE 1.985%), outperforming ARIMA(0,1,0) (MAPE 4.336%), NNAR(5,9) (4.340%), SARIMA(0,1,0)(0,0,1)[20] (4.350%), and Random Forest (4.466%). Overall, the findings suggest that the CNY/IDR series is driven mainly by a strong upward trend, which smoothing-based methods capture more effectively than the statistical, machine-learning, or neural-network alternatives.

Keywords—exchange rate forecasting; CNY/IDR; ARIMA; Seasonal ARIMA; Double Moving Average; Random Forest; Neural Network Autoregression; time series.

I. INTRODUCTION

A. Background

The exchange rate is one of the most closely monitored macroeconomic variables, as its movements directly affect international trade, investment, inflation, and the conduct of monetary policy. Because exchange rates are highly sensitive to economic and political uncertainty, the ability to anticipate their future values is important for central banks, exporters, importers, and firms managing currency risk [19].

In the Indonesian context, the importance of the Chinese yuan (CNY) relative to the rupiah (IDR) has grown rapidly

alongside the deepening of economic ties between Indonesia and China. Bilateral trade between the two countries surpassed the US\$150 billion milestone in 2025, reaching approximately US\$167.49 billion, while the bilateral currency swap arrangement was renewed in January 2025 and expanded to 400 billion yuan, equivalent to about US\$59 billion [21]. A particularly significant structural change is the shift away from settling transactions through the United States dollar: the China-Indonesia Local Currency Settlement mechanism launched in 2021 was upgraded in 2025 to the Local Currency Transaction framework, expanding local-currency settlement to all balance-of-payments items [21]. In the first seven months of 2025, such transactions between the two countries reached US\$6.23 billion, around 45% of Indonesia's total local-currency transaction volume with all partners [12]. As a growing share of trade is settled directly in yuan and rupiah, the CNY/IDR rate increasingly determines transaction costs and competitiveness for businesses in both countries.

Against this background, accurate forecasting of the CNY/IDR exchange rate has become increasingly relevant. Reliable forecasts support hedging decisions, trade planning, and policy formulation, and help economic actors prepare for adverse currency movements before they occur.

B. Problem Statement

Despite its rising importance, the CNY/IDR exchange rate is difficult to predict. By April 2026 the rupiah had weakened by about 4.9% year-to-date against the yuan, trading at roughly Rp2,505 per CNY. This decline was steeper than the rupiah's movement against the US dollar over the same period, and it was driven partly by surging imports from China [7]. Such pronounced trends, together with short-term fluctuations, make the series hard to model. The available forecasting methods also rest on different assumptions: statistical models such as ARIMA and its seasonal extension SARIMA assume linear dependence, with SARIMA additionally accommodating periodic seasonal structure; smoothing methods such as the Double Moving Average are built for trended series; and machine-learning and neural-

network methods such as Random Forest and NNAR can capture nonlinear behavior. Since no single method works best in every setting, the most suitable model for this particular pair has to be found empirically. The existing Indonesian forecasting literature, however, concentrates almost entirely on the IDR/USD pair [1], [2], [4], leaving the CNY/IDR pair largely unexamined.

C. Research Objective

This study pursues two objectives. The first is to compare how well five approaches, ARIMA, Seasonal ARIMA (SARIMA), Double Moving Average, Random Forest, and Neural Network Autoregression, forecast the daily CNY/IDR exchange rate. The second is to identify which of them predicts the pair most accurately, judged by out-of-sample evaluation.

D. Research Contribution

This study makes three main contributions. First, it applies five different forecasting approaches to the daily CNY/IDR exchange rate. Second, it places statistical (ARIMA and SARIMA), smoothing (DMA), machine-learning (Random Forest), and neural-network (NNAR) models side by side on a single dataset. Third, by ranking them with RMSE, MAE, and MAPE, it offers empirical evidence on a currency pair that is economically important yet has received little research attention

II. LITERATURE REVIEW

A. Time Series Forecasting

Time series forecasting uses historical, time-ordered observations to predict future values of a variable, on the assumption that patterns embedded in past data, such as trend, seasonality, and autocorrelation, carry information useful for prediction [13]. Forecasting methods are commonly grouped into statistical approaches, which model linear dependence explicitly, and data-driven approaches such as machine learning and neural networks, which can approximate nonlinear relationships. Exchange rates are a common application, since they are recorded frequently and carry clear economic weight, but they are also noisy and easily disturbed by external shocks, which is why the choice of forecasting method matters [11], [19].

B. Autoregressive Integrated Moving Average (ARIMA)

ARIMA is among the most widely used statistical methods for time series forecasting [2]. It combines three components. The autoregressive (AR) component of order p expresses the current value as a linear function of its own past values; the moving-average (MA) component of order q expresses it as a linear function of past forecast errors; and the integrated component of order d applies differencing to remove trend and achieve stationarity. A series must be stationary in both variance and mean before fitting. Mean stationarity is typically examined with the Augmented Dickey-Fuller (ADF) test, and differencing is applied until stationarity is achieved. The general ARIMA(p, d, q) model can be written as

$$\varphi_p(B)(1 - B)^d y_t = \theta_q(B) \varepsilon_t \quad (1)$$

where B is the backshift operator, $\varphi(B)$ and $\theta(B)$ are the AR and MA polynomials, and ε is white noise. After

estimation, candidate models are evaluated through residual diagnostics, in particular tests for normality and for the absence of autocorrelation (white noise). ARIMA has been applied extensively to the rupiah against the US dollar [1], [2], [4] and serves here as the statistical benchmark, although comparative studies show that it does not always outperform machine-learning or neural-network alternatives [11], [14], [19].

C. Seasonal ARIMA (SARIMA)

Seasonal ARIMA (SARIMA) extends the ARIMA model by adding seasonal autoregressive, differencing, and moving-average terms to capture periodic patterns in a time series [17]. It is denoted SARIMA(p, d, q)(P, D, Q)[s], where the lowercase terms describe the non-seasonal structure and the uppercase terms (P, D, Q) describe the seasonal structure repeating every s periods. The seasonal component is intended to model recurring fluctuations that the non-seasonal ARIMA model cannot represent. As with ARIMA, candidate SARIMA models are estimated and then evaluated through parameter significance testing and residual diagnostics for normality and white noise. SARIMA is included here to examine whether the CNY/IDR exchange rate series contains exploitable seasonal structure beyond its trend behavior.

D. Double Moving Average (DMA)

The Double Moving Average method is a smoothing technique suited to data that exhibit a trend but no seasonality [9]. It applies a moving average to the original series to obtain a first moving average M_1 , then a second moving average to obtain M_2 . The first moving average smooths short-term fluctuations, while the difference between the two captures the trend component. The estimated level a_t , trend b_t , and h -step-ahead forecast are computed as

$$a_t = 2M_1 - M_2 \quad (2)$$

$$b_t = [2 / (n - 1)] (M_1 - M_2) \quad (3)$$

$$F_{t+h} = a_t + b_t h \quad (4)$$

where n is the moving-average window length and h the forecast horizon. Because performance depends on the window size, several windows are typically evaluated. Empirical studies report that DMA can achieve very low forecasting errors on strongly trended data [10].

E. Random Forest

Random Forest is an ensemble machine-learning method that combines many decision trees [3]. Each tree is grown on a bootstrap sample of the training data (bootstrap aggregation, or bagging), and a random subset of features is considered at each split, which reduces correlation among trees and lowers variance; the regression prediction is the average over all trees. For time series forecasting, the univariate series is transformed into a supervised-learning problem in which lagged values $y_{t-1}, \dots, y_{t-p}$ serve as predictors of y_t [16]. Random Forest also yields a variable-importance measure indicating which lags contribute most. Prior work shows that it can outperform classical models in exchange-rate forecasting [3], and that its accuracy is generally highest when only a small number of recent lags is used [16].

F. Neural Network Autoregression (NNAR)

NNAR applies a feed-forward neural network to time series forecasting by using lagged observations as inputs [13]. In its non-seasonal form it is denoted NNAR(p,k), where p is the number of lagged inputs and k the number of nodes in a single hidden layer. An NNAR($p,0$) model is equivalent to an ARIMA($p,0,0$) model but without the stationarity restrictions, while the hidden layer allows the network to approximate nonlinear relationships. To reduce sensitivity to random weight initialization, several networks are trained and their forecasts averaged. Empirical results are mixed: NNAR has outperformed ARIMA in some applications [8] but been outperformed in others [5], indicating that its advantage depends on whether the series contains exploitable nonlinear structure.

G. Evaluation Metrics

Forecasting performance is assessed on the testing set using three standard error metrics. The Root Mean Square Error (RMSE) penalizes large errors more heavily; the Mean Absolute Error (MAE) measures the average absolute deviation; and the Mean Absolute Percentage Error (MAPE) expresses the average error as a scale-independent percentage:

$$RMSE = \sqrt{(1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$MAE = (1/n) \sum_{i=1}^n |y_i - \hat{y}_i| \quad (6)$$

$$MAPE = (100/n) \sum_{i=1}^n |(y_i - \hat{y}_i) / y_i| \quad (7)$$

where y_i is the actual value, $\hat{y}_i$ the predicted value, and n the number of testing observations. Lower values indicate better accuracy.

III. METHODOLOGY

A. Dataset

The dataset used in this research consists of daily CNY (China Yuan) to IDR (Indonesia Rupiah) exchange rate data collected from Investing.com. The observation period spans from 1 January 2024 to 29 May 2026, resulting in 630 observations. The weekend and public holiday are excluded due to the absence of trading activities. The exchange rate value used in this study is the closing exchange rate. The dataset summary is given in Table I.

TABLE I. DATASET SUMMARY

Item	Value
Currency pair	CNY/IDR
Source	Investing.com
Period	Jan 2024 to May 2026
Observations	630
Value used	Daily closing rate

B. Data Preprocessing

For model development, several preprocessing steps were performed to ensure the suitability of the dataset for time series forecasting. First, the exchange rate were converted into a numerical format by removing thousand separators and standardizing decimal notation. Next, the cleaned exchange rate values were transformed into a time series object to support further time series analysis and forecasting procedures.

Descriptive statistical analysis was conducted to examine the characteristics of the dataset, including the minimum, maximum, mean, median, and quartile values. In addition, a time series plot was generated to provide an initial understanding of the trend and fluctuation patterns of the CNY/IDR exchange rate throughout the observation period.

C. Stationarity Testing

Stationarity testing was conducted before model development. Variance stationarity was evaluated using the Power Transformation approach. The estimated transformation parameter (λ) and the likelihood ratio test were used to determine whether a variance-stabilizing transformation was required [17]. Additionally, Levene's Test was employed as a supporting procedure to assess variance homogeneity because it is relatively robust to departures from normality and less sensitive to outliers [18].

Mean stationarity was then examined using the Augmented Dickey-Fuller (ADF) test. The null hypothesis of the ADF test states that the series contains a unit root and is therefore non-stationary, whereas the alternative hypothesis indicates that the series is stationary [15]. If the ADF test failed to reject the null hypothesis at the 5% significance level, differencing was applied until stationarity was achieved.

D. Train-Test Split

To evaluate forecasting performance objectively, the dataset was divided into training and testing sets using an 80:20 ratio. The training set was used for model development, whereas the testing set was reserved for out-of-sample evaluation.

Since the dataset represents a time series, the split was performed chronologically to preserve the temporal order of observations and avoid information leakage. Consequently, the first 504 observations were allocated to the training set, while the remaining 126 observations were assigned to the testing set.

E. Forecasting Models

1) ARIMA

The Autoregressive Integrated Moving Average (ARIMA) model was employed as the statistical forecasting approach. Candidate models were identified using the ACF and PACF plots after stationarity was achieved [20]. Several ARIMA specifications were estimated and compared to determine the most suitable model.

2) Seasonal ARIMA (SARIMA)

Seasonal ARIMA (SARIMA) was employed to examine whether the exchange rate series contained seasonal patterns beyond its trend behavior. Candidate seasonal specifications were estimated by extending the ARIMA structure with seasonal autoregressive, differencing, and moving-average terms, and were then compared to identify the most suitable model.

3) Double Moving Average (DMA)

Double Moving Average (DMA) was utilized to capture the trend component of the exchange rate series.

The model applies moving average smoothing twice to estimate both level and trend components [9]. Several window sizes were evaluated to identify the most accurate DMA configuration.

4) Random Forest

Random Forest was implemented as a machine learning forecasting model. Lagged exchange rate values were generated as predictor variables, and the model was trained using an ensemble of decision trees. The final prediction was obtained by averaging the outputs of all trees

5) Neural Network Autoregression (NNAR)

Neural Network Autoregression (NNAR) was employed as a deep learning-based forecasting approach. The model utilizes lagged observations as inputs to a feed-forward neural network and automatically determines the network structure based on the characteristics of the training data.

F. Model Evaluation

The forecasting performance of all models was evaluated using RMSE, MAE, and MAPE [6]. These metrics were calculated on the testing dataset to compare the predictive accuracy of the proposed models.

IV. RESULTS AND DISCUSSION

A. Exploratory Data Analysis

Exploratory Data Analysis was conducted to provide an overview of the CNY/IDR exchange rate dataset. The summary statistics including the minimum, maximum, mean, median, and quartile values are presented in Table II.

TABLE II. DESCRIPTIVE STATISTICS OF THE CNY/IDR SERIES

Min.	Q1	Median	Mean	Q3	Max.
2146	2205	2260	2285	2338	2641

As shown in Table II, the CNY/IDR exchange rate ranged from 2,146 IDR per CNY to 2,641 IDR per CNY during the observation period. The average exchange rate was 2,285 IDR per CNY, while the median value was 2,260 IDR per CNY. The relatively small difference between the mean and median indicates that the data distribution is reasonably balanced.

To visualize the overall behavior of the exchange rate, a time series plot was generated, as shown in Fig. 1.

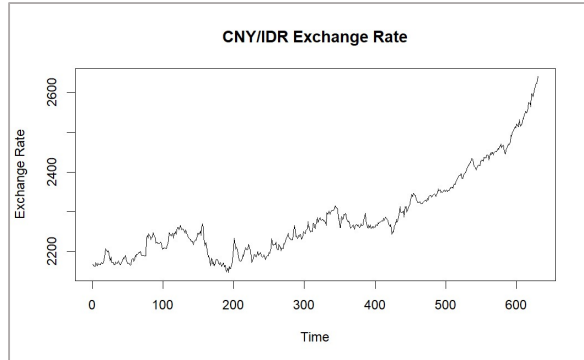
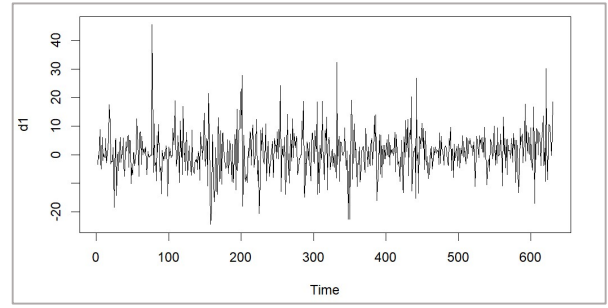


FIG. 1. TIME SERIES PLOT OF THE DAILY CNY/IDR EXCHANGE RATE

Based on Fig. 1, the exchange rate exhibits a clear upward trend, increasing from approximately 2,167 IDR per CNY in



January 2024 to around 2,641 IDR per CNY in May 2026. Although short-term fluctuations are present throughout the observation period, the overall trend indicates a continuous appreciation of the Chinese Yuan against the Indonesian Rupiah. This pattern suggests that the series may be non-stationary and requires further stationarity testing before model development.

B. Stationarity Testing

Stationarity testing was performed to ensure that the time series met the assumptions required by forecasting models, particularly ARIMA.

1) Variance Stationarity

Variance stationarity was evaluated using the Power Transformation approach. The estimated transformation parameter and likelihood ratio test results are presented in Table III.

TABLE III. VARIANCE STATIONARITY (POWER TRANSFORMATION)

Parameter	Estimated Lambda	LRT Statistic	p-value
Value	-2.9996	106.5809	< 2.22e-16

The likelihood ratio test produced a p-value smaller than 0.05, indicating that the null hypothesis of variance stationarity was rejected. Therefore, the series was initially identified as non-stationary with respect to variance.

However, the estimation process generated a convergence warning, indicating that the estimated transformation parameter might not be reliable. To further assess variance stability, Levene's Test was conducted. The results are presented in Table IV.

TABLE IV. LEVENE'S TEST RESULT

Statistic	F-value	p-value
Value	0.0122	0.9982

The Levene's Test yielded a p-value greater than 0.05, indicating no significant evidence of variance heterogeneity. Therefore, the variance of the series was considered sufficiently stable for forecasting purposes, and no variance transformation was applied.

2) Mean Stationarity

Mean stationarity was examined using the Augmented Dickey-Fuller (ADF) test. The ADF test produced a p-value greater than 0.05, indicating that the null hypothesis could not be rejected. Therefore, the original CNY/IDR exchange rate series was non-stationary in mean. To achieve stationarity, first-order differencing was applied. The resulting differenced series is illustrated in Fig. 2, and the ADF test was then repeated

on the differenced series. The results are shown in Table V.

TABLE V. ADF TEST BEFORE AND AFTER DIFFERENCING

Series	DF statistic	p-value	Result
Original	0.51995	0.99	Non-stationary
First diff.	-9.1132	0.01	Stationary

FIG. 2. FIRST-ORDER DIFFERENCED CNY/IDR EXCHANGE RATE SERIES

The differenced series produced a p-value below 0.05, indicating that the null hypothesis was rejected. Therefore, the series became stationary after first-order differencing. Consequently, the order of differencing for ARIMA modeling was set to $d = 1$.

C. ARIMA Model

1) Model Identification

To identify suitable ARIMA structures, the ACF and PACF plots of the differenced series were examined, as shown in Fig. 3 and Fig. 4.

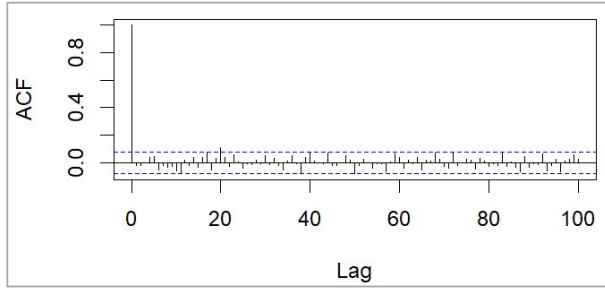


FIG. 3. ACF PLOT OF THE FIRST-DIFFERENCED SERIES

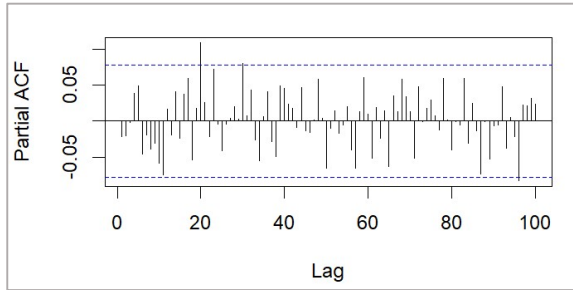


FIG. 4. PACF PLOT OF THE FIRST-DIFFERENCED SERIES

The ACF plot shows that most autocorrelations fall within the significance bounds, while the PACF plot exhibits only a few significant spikes. Based on these observations, several candidate ARIMA models were considered, including ARIMA(0,1,0), ARIMA(0,1,1), ARIMA(1,1,0), ARIMA(1,1,2), and ARIMA(2,1,0).

The estimated parameters of the candidate models were subsequently evaluated using significance testing, diagnostic checking, and forecasting performance measures.

2) Parameter Estimation and Diagnostic Checking

The significance of the estimated parameters was evaluated using the z-test. The results are presented in Table VI.

TABLE VI. PARAMETER SIGNIFICANCE TEST RESULTS

Model	Significant Parameters
ARIMA(0,1,1)	None
ARIMA(1,1,0)	None
ARIMA(1,1,2)	None
ARIMA(2,1,0)	None

ARIMA(0,1,0) does not contain autoregressive (AR) or moving average (MA) parameters. Therefore, parameter significance testing was not applicable for this model, and the model was directly evaluated using residual diagnostic tests and forecasting performance measures.

For the remaining ARIMA models, parameter significance was assessed using the z-test. The results indicated that all estimated AR and MA coefficients produced p-values greater than 0.05. Consequently, none of the parameters were statistically significant at the 5% significance level.

Residual diagnostics were subsequently performed using the Lilliefors normality test and the Box-Pierce white-noise test. The results are summarized in Table VII.

TABLE VII. RESIDUAL DIAGNOSTIC RESULTS OF ARIMA MODELS

Model	Normality	White Noise
ARIMA(0,1,0)	Not Normal	Satisfied
ARIMA(0,1,1)	Not Normal	Satisfied
ARIMA(1,1,0)	Not Normal	Satisfied
ARIMA(1,1,2)	Not Normal	Satisfied
ARIMA(2,1,0)	Not Normal	Satisfied

The Lilliefors test indicated that the residuals of all candidate ARIMA models were not normally distributed. However, the Box-Pierce test produced p-values greater than 0.05, indicating that the residuals behaved as white noise and contained no significant autocorrelation.

Although the ARIMA models did not satisfy the residual normality assumption and the estimated parameters were not statistically significant, they were retained for forecasting evaluation because model performance was assessed using an out-of-sample testing dataset. Among the candidate models, ARIMA(0,1,0) achieved the lowest forecasting errors and was therefore selected as the representative ARIMA model for comparison with the other forecasting approaches.

3) Forecasting Performance

The forecasting performance of the ARIMA models was evaluated using the testing dataset. The results are shown in Table VIII.

TABLE VIII. FORECASTING PERFORMANCE OF ARIMA MODELS

Model	RMSE	MAE	MAPE (%)
ARIMA(0,1,0)	128.639	108.585	4.336
ARIMA(0,1,1)	128.649	108.597	4.337
ARIMA(1,1,0)	128.648	108.596	4.337
ARIMA(1,1,2)	128.712	108.671	4.340
ARIMA(2,1,0)	128.713	108.671	4.340

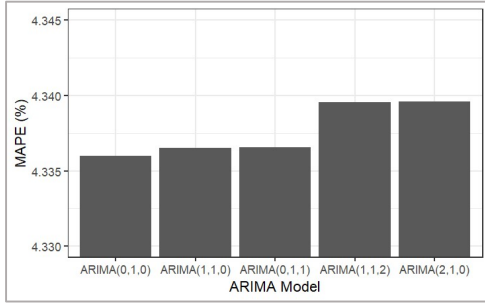


FIG. 5. COMPARISON OF MAPE VALUES ACROSS ARIMA MODELS

As shown in Fig. 5, the forecasting performance of the candidate ARIMA models was highly similar, with MAPE values ranging only from 4.336% to 4.340%. The differences between the models were therefore negligible, indicating that increasing model complexity did not provide meaningful improvements in forecasting accuracy.

This finding is consistent with the parameter significance tests, where none of the estimated AR and MA coefficients were statistically significant. Consequently, the differenced CNY/IDR exchange rate series behaved similarly to a random walk process, making the simpler ARIMA(0,1,0) model preferable to more complex ARIMA structures.

Based on Table VIII and Fig. 5, ARIMA(0,1,0) produced the lowest RMSE, MAE, and MAPE values. Therefore, ARIMA(0,1,0) was selected as the representative ARIMA model for comparison with the other forecasting approaches.

D. SARIMA Model

1) Model Identification

Prior stationarity testing showed that the original CNY/IDR exchange rate series was non-stationary in mean and became stationary after first-order differencing. Therefore, the differencing order was fixed at $d=1$.

The ACF and PACF plots of the first-differenced series shown previously in Fig. 3 and Fig. 4 were further examined to investigate possible seasonal behavior. Although most autocorrelations were within the significance bounds, noticeable spikes were observed around lag 20 in both plots. These patterns indicate the presence of a weak seasonal structure with an approximate period of 20 observations.

To further explore this seasonal behavior, a decomposition analysis was performed using a seasonal period of 20 observations. The decomposition plot revealed a recurring seasonal component together with an upward trend in the exchange rate series.

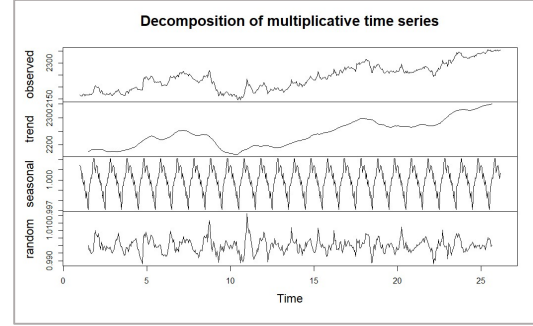


FIG. 6. DECOMPOSITION PLOT OF THE CNY/IDR EXCHANGE RATE SERIES

Based on these findings, several SARIMA candidate models incorporating seasonal autoregressive and seasonal moving-average components with a seasonal period of 20 observations were considered:

- SARIMA(0,1,0)(1,0,0)[20]
- SARIMA(0,1,1)(1,0,0)[20]
- SARIMA(1,1,0)(1,0,0)[20]
- SARIMA(0,1,0)(0,0,1)[20]

The candidate models were subsequently evaluated through parameter significance testing, residual diagnostics, and forecasting performance measures.

2) Parameter Estimation and Diagnostic Checking

The significance of the estimated parameters was evaluated using the z-test. The results are summarized in Table IX.

TABLE IX. PARAMETER SIGNIFICANCE TEST RESULTS

Model	Significant Parameters
SARIMA(0,1,0)(1,0,0)[20]	None
SARIMA(0,1,1)(1,0,0)[20]	None
SARIMA(1,1,0)(1,0,0)[20]	AR(1), SAR(1)
SARIMA(0,1,0)(0,0,1)[20]	None

The z-test results indicate that only SARIMA(1,1,0)(1,0,0)[20] contained statistically significant parameters at the 5% significance level. Both the non-seasonal autoregressive parameter and the seasonal autoregressive parameter were significant. The remaining candidate models did not contain significant seasonal or non-seasonal coefficients. These results suggest that the seasonal component contributed only limited explanatory power to the CNY/IDR exchange rate series.

Residual diagnostics were subsequently performed using the Lilliefors normality test and the Box-Pierce white-noise test. The results are summarized in Table X.

TABLE X. RESIDUAL DIAGNOSTIC RESULTS OF SARIMA MODELS

Model	Normality	White Noise
SARIMA(0,1,0)(1,0,0)[20]	Not Normal	Satisfied
SARIMA(0,1,1)(1,0,0)[20]	Not Normal	Satisfied
SARIMA(1,1,0)(1,0,0)[20]	Not Normal	Satisfied
SARIMA(0,1,0)(0,0,1)[20]	Not Normal	Satisfied

The Lilliefors test produced p-values below 0.05 for all candidate SARIMA models, indicating that the

residuals were not normally distributed. However, the Box-Pierce test yielded p-values greater than 0.05, suggesting that the residuals behaved as white noise and contained no significant autocorrelation.

Although the residual normality assumption was not satisfied, all candidate SARIMA models fulfilled the white-noise assumption and were therefore retained for forecasting evaluation.

3) Forecasting Performance

The forecasting performance of the SARIMA models was evaluated using the testing dataset. The results are presented in Table XI.

TABLE XI. FORECASTING PERFORMANCE OF SARIMA MODELS

Model	RMSE	MAE	MAPE (%)
SARIMA(0,1,0)(1,0,0)[20]	128.965	108.954	4.351
SARIMA(0,1,1)(1,0,0)[20]	128.967	108.957	4.351
SARIMA(1,1,0)(1,0,0)[20]	128.964	108.953	4.351
SARIMA(0,1,0)(0,0,1)[20]	128.936	108.919	4.350

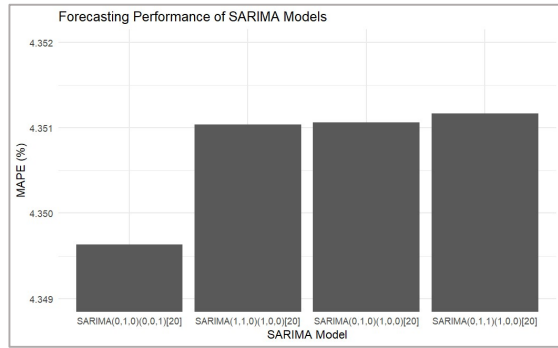


FIG. 7. COMPARISON OF SARIMA MODEL PERFORMANCE BASED ON MAPE

Based on Table XI, SARIMA(0,1,0)(0,0,1)[20] achieved the lowest RMSE, MAE, and MAPE values among the evaluated SARIMA models, with an RMSE of 128.936, an MAE of 108.919, and a MAPE of 4.350%.

However, the differences in forecasting performance among the candidate SARIMA models were relatively small. Furthermore, the best-performing SARIMA model did not contain statistically significant seasonal parameters. Although SARIMA models were developed to capture possible seasonal behavior in the exchange rate series, the forecasting results indicate that incorporating seasonal components did not provide substantial improvements in predictive accuracy.

Compared with the non-seasonal ARIMA(0,1,0) model, which achieved a MAPE of 4.336%, the best SARIMA model produced a slightly higher MAPE of 4.350%. This finding suggests that seasonal effects were relatively weak and contributed little additional predictive information for forecasting the CNY/IDR exchange rate during the observation period.

Therefore, although SARIMA models were evaluated as an extension of ARIMA, the non-seasonal ARIMA(0,1,0) model remained the preferred statistical forecasting approach in this study due to its slightly better forecasting performance and simpler model structure.

E. Double Moving Average (DMA) Model

1) Model Development

Three DMA configurations were evaluated using moving-average windows of 5, 10, 20, 23, 24, 25, 26 and 30 observations. The first moving average was used to smooth short-term fluctuations, while the second moving average was used to estimate the trend component.

2) Forecasting Performance

The forecasting results are presented in Table IX.

TABLE IX. FORECASTING PERFORMANCE OF DMA MODELS

Model	RMSE	MAE	MAPE (%)
DMA(5)	111.808	93.790	3.744
DMA(10)	115.905	97.580	3.896
DMA(20)	71.917	57.361	2.284
DMA(23)	64.382	50.259	1.999
DMA(24)	64.025	49.929	1.985
DMA(25)	65.037	50.891	2.024
DMA(26)	65.684	51.483	2.048
DMA(30)	72.257	57.679	2.297

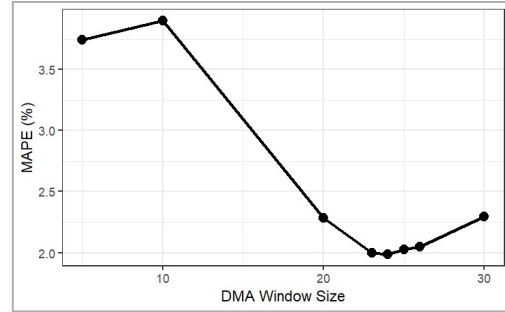


FIG. 8. COMPARISON OF DMA MODEL PERFORMANCE BASED ON MAPE

As shown in Fig. 8, the forecasting accuracy improved substantially as the moving-average window increased from 5 to 24 observations. The MAPE value decreased from 3.744% for DMA(5) to 1.985% for DMA(24). However, further increases in the window size beyond 24 observations resulted in higher forecasting errors.

This finding indicates that the selection of the smoothing window has a significant impact on forecasting performance. A window size that is too small may be unable to adequately smooth short-term fluctuations, whereas an excessively large window may oversmooth the series and reduce responsiveness to recent exchange rate movements.

Among the evaluated DMA models, DMA(24) achieved the best forecasting performance. The superior performance of DMA(24) suggests that it was able to effectively smooth short-term fluctuations while preserving the underlying trend of the CNY/IDR exchange rate series, resulting in more accurate forecasts.

F. Random Forest Model

1) Model Development

To identify the most suitable Random Forest configuration, several combinations of the number of

trees (ntree) and the number of variables randomly selected at each split (mtry) were evaluated.

2) Forecasting Performance

The forecasting performance of each configuration was assessed using RMSE, MAE, and MAPE on the testing dataset. The results are presented in Table X.

TABLE X. FORECASTING PERFORMANCE OF RANDOM FOREST

ntree	mtry	RMSE	MAE	MAPE (%)
25	1	131.480	112.160	4.481
25	2	133.026	113.951	4.553
30	1	131.310	111.968	4.473
30	2	132.783	113.671	4.542
35	1	131.211	111.855	4.468
35	2	132.660	113.531	4.536
36	1	131.176	111.813	4.466
36	2	132.540	113.393	4.531
37	1	131.271	111.925	4.471
37	2	132.547	113.400	4.531
40	1	131.327	111.991	4.474
40	2	132.493	113.338	4.529
50	1	131.468	112.156	4.481
50	2	132.573	113.426	4.532

The results indicate that models with $mtry = 1$ consistently outperformed those with $mtry = 2$ across all evaluation metrics. Among the evaluated configurations, the Random Forest model with $ntree = 36$ and $mtry = 1$ achieved the best forecasting performance, yielding an RMSE of 131.176, an MAE of 111.813, and a MAPE of 4.466%. Fig. 8 presents the forecasting performance of the evaluated Random Forest configurations.

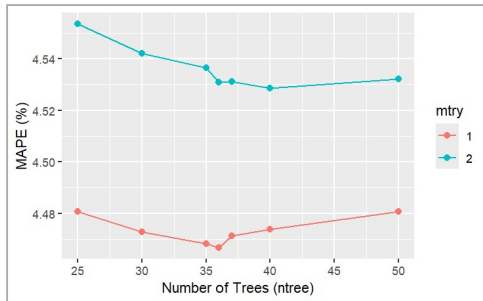


FIG. 9. COMPARISON OF MAPE VALUES ACROSS RANDOM FOREST CONFIGURATIONS

As shown in Fig. 9, the forecasting performance was relatively stable across different parameter combinations, indicating that the Random Forest model was not highly sensitive to changes in the number of trees. However, configurations with $mtry = 1$ consistently produced lower forecasting errors than those with $mtry = 2$. This suggests that using a smaller subset of predictors at each split improved model generalization and reduced overfitting.

Based on the overall evaluation results, the Random Forest model with $ntree = 36$ and $mtry = 1$ achieved the best forecasting performance, yielding an RMSE of 131.176, an MAE of 111.813, and a MAPE of 4.466%. Therefore, this configuration was selected as the final

Random Forest model for subsequent analysis and comparison with the other forecasting approaches.

3) Variable Importance Analysis

The importance of predictor variables was evaluated using the Mean Decrease in Accuracy (%IncMSE) and IncNodePurity measures. The results are presented in Fig. 10.

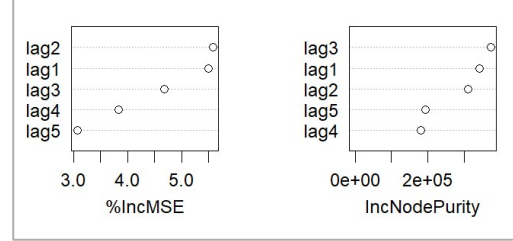


FIG. 10. VARIABLE IMPORTANCE PLOT OF THE SELECTED RANDOM FOREST MODEL

Based on the %IncMSE metric, lag2 was identified as the most influential predictor, followed by lag1 and lag3. This result indicates that the exchange rate values observed during the previous two trading days contributed the most to forecasting accuracy. Meanwhile, the IncNodePurity measure shows that lag3 contributed the most to the tree-splitting process, followed by lag1 and lag2.

Although the ranking of predictor importance differs slightly between the two measures, both indicate that the most recent observations (lag1, lag2, and lag3) contain the most relevant information for forecasting future CNY/IDR exchange rate movements. Therefore, short-term historical exchange rate values play a dominant role in the Random Forest forecasting model.

G. Neural Network Regression

1) Model Development

To determine the most suitable Neural Network Autoregression (NNAR) architecture, several combinations of input lags (p) and hidden neurons (size) were evaluated. The forecasting performance of each configuration was assessed using RMSE, MAE, and MAPE on the testing dataset.

2) Forecasting Performance

The forecasting performance of the evaluated NNAR models is presented in Table XI.

TABLE XI. FORECASTING PERFORMANCE OF NNAR MODELS

p	size	RMSE	MAE	MAPE (%)
4	7	132.911	113.460	4.534
4	10	133.236	113.833	4.549
4	9	133.270	113.877	4.551
4	8	133.941	114.608	4.580
5	7	129.677	109.717	4.382
5	8	129.586	109.655	4.380
5	9	128.751	108.689	4.340
5	10	129.802	109.890	4.389
6	7	132.081	112.491	4.495
6	8	131.140	111.402	4.450
6	9	131.504	111.836	4.468
6	10	130.163	110.290	4.405

The results indicate that the forecasting accuracy of the NNAR model was influenced by both the number of lagged observations and the number of hidden neurons. In general, models with a larger number of lagged inputs achieved lower forecasting errors than simpler architectures. This finding suggests that the CNY/IDR exchange rate series contains temporal dependencies extending beyond the most recent observation.

Among the evaluated configurations, the NNAR(5,9) model achieved the best forecasting performance, yielding an RMSE of 128.751, an MAE of 108.689, and a MAPE of 4.340%. Fig. X presents the comparison of MAPE values across the evaluated NNAR architectures.

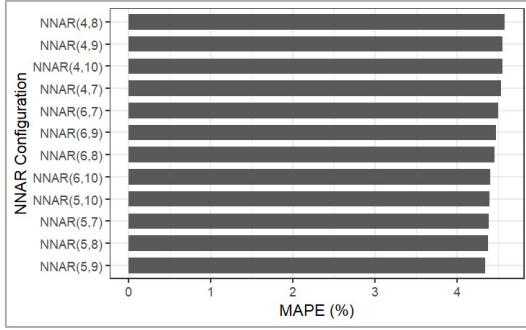


FIG.11 COMPARISON OF MAPE VALUES ACROSS NNAR CONFIGURATIONS

As shown in Fig. 11, the forecasting performance generally improved when the number of lagged inputs increased from four to five. However, further increases in model complexity did not consistently improve forecasting accuracy. The NNAR(5,9) architecture provided the best balance between model complexity and predictive performance, while architectures with fewer lagged inputs or excessively large hidden layers tended to produce higher forecasting errors. Therefore, NNAR(5,9) was selected as the final neural network model for comparison with the other forecasting approaches.

H. Comparison of Forecasting Models

TABLE XII. COMPARISON OF FORECASTING MODEL PERFORMANCE

Model	RMSE	MAE	MAPE (%)
ARIMA(0,1,0)	128.639	108.585	4.336
SARIMA(0,1,0)(0,0,1)[20]	128.936	108.919	4.350
DMA(24)	64.025	49.929	1.985
NNAR(5,9)	128.751	108.689	4.340
RF(36,1)	131.176	111.813	4.466

Table XII presents the forecasting performance comparison among the selected forecasting models. Based on the evaluation results, DMA(24) consistently achieved the lowest RMSE, MAE, and MAPE values among all evaluated approaches. Specifically, DMA(24) produced an RMSE of 64.025, an MAE of 49.929, and a MAPE of 1.985%, indicating substantially higher forecasting accuracy than the competing models.

Among the remaining approaches, ARIMA(0,1,0) achieved the best performance with an RMSE of 128.639, an MAE of 108.585, and a MAPE of 4.336%. The NNAR(5,9) model produced very similar results, yielding an RMSE of 128.751, an MAE of 108.689, and a MAPE

of 4.340%. Likewise, SARIMA(0,1,0)(0,0,1)[20] generated comparable forecasting performance with an RMSE of 128.936, an MAE of 108.919, and a MAPE of 4.350%. The small differences among these three models suggest that incorporating seasonal components or nonlinear neural-network structures did not provide substantial forecasting improvements over the simpler ARIMA model.

The Random Forest model with $n_{tree}=36$ and $m_{try}=1$ produced the highest forecasting errors among the evaluated approaches, with an RMSE of 131.176, an MAE of 111.813, and a MAPE of 4.466%. Although Random Forest is capable of capturing nonlinear relationships through ensemble learning, the lag-based predictor variables were insufficient to fully represent the underlying dynamics of the CNY/IDR exchange rate series.

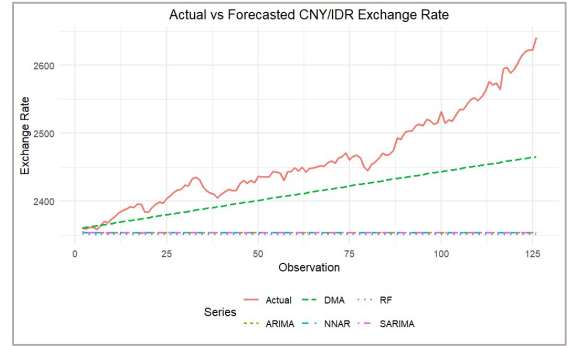


FIG.12 ACTUAL AND FORECASTED VALUES OF THE SELECTED FORECASTING MODELS

As illustrated in Fig. 12, the forecasts generated by ARIMA, SARIMA, NNAR, and Random Forest remained relatively stable throughout the testing period and were unable to adequately follow the upward movement of the actual exchange rate. In contrast, DMA(24) successfully captured the increasing trend of the series and produced forecasts that more closely followed the observed values. This ability to preserve the underlying trend while smoothing short-term fluctuations contributed to its substantially lower forecasting errors.

Overall, DMA(24) outperformed the statistical, machine-learning, and neural-network approaches across all evaluation metrics. Therefore, DMA(24) was selected as the best forecasting model for predicting the daily CNY/IDR exchange rate in this study.

V. CONCLUSION

This study compared the forecasting performance of five forecasting approaches, namely ARIMA, Seasonal ARIMA (SARIMA), Double Moving Average (DMA), Random Forest, and Neural Network Autoregression (NNAR), for predicting the daily CNY/IDR exchange rate. Using 630 daily observations collected from January 2024 to May 2026, the dataset was divided into training and testing sets using an 80:20 ratio. Prior to model development, stationarity testing indicated that the original series was non-stationary in mean and became stationary after first-order differencing.

The forecasting results demonstrated that DMA(24) achieved the highest predictive accuracy among all evaluated models, with an RMSE of 64.025, an MAE of 49.929, and a MAPE of 1.985%. Among the remaining approaches, ARIMA(0,1,0), NNAR(5,9), and SARIMA(0,1,0)(0,0,1)[20] produced similar forecasting performance, with MAPE values of 4.336%, 4.340%, and 4.350%, respectively. Meanwhile, the Random Forest model (ntree=36, mtry=1) yielded the largest forecasting errors, with a MAPE of 4.466%.

The comparison of forecasted and actual values further revealed that DMA(24) was better able to follow the upward trend observed in the testing period. In contrast, ARIMA, SARIMA, NNAR, and Random Forest tended to generate relatively stable forecasts that were less responsive to the increasing movement of the exchange rate. These findings indicate that the CNY/IDR exchange rate series during the study period was primarily characterized by a strong trend component rather than pronounced seasonal or nonlinear patterns.

Overall, the results suggest that smoothing-based forecasting methods are more suitable for the studied exchange rate series than the statistical, machine-learning, and neural-network approaches considered in this study. Therefore, DMA(24) was identified as the most appropriate forecasting model for predicting the daily CNY/IDR exchange rate during the observation period. Future research may incorporate additional macroeconomic variables, longer observation periods, or hybrid forecasting approaches to further improve forecasting performance.

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